

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models
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Submission Date	31/8/2026

1 Objective

To understand and implement ensemble learning strategies including Bagging, Boosting, and Stacking. To compare these models in terms of accuracy, stability, and generalization, and analyze their effect on bias and variance.

2 Problem Statement

The problem involves classifying tumor samples from the Wisconsin Diagnostic Breast Cancer dataset. We employ heterogeneous ensemble models to combine multiple base learners and theoretically compare how Bagging reduces variance while Boosting addresses model bias.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30
Number of Classes	2 (Malignant, Benign)
Missing Values	None
Train-Test Split	80-20

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

Figure Template

Inference:

Exploratory Data Analysis was thoroughly conducted in the previous experiment. The data distributions and correlations remain identical, so we proceed directly to modeling.

5 Data Preprocessing

Explain:

- Missing value handling
- Encoding
- Feature scaling
- Train-test split
- Feature engineering (if applicable)

6 Machine Learning Algorithm

Explain:

- Working principle
- Mathematical formulation
- Advantages
- Limitations

7 Experimental Setup

Python Version
Scikit-Learn Version
IDE
Operating System
Hardware

8 Hyperparameter Selection

Parameter	Value
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9 Results

Metric	Value
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Table 4: Performance Comparison of Ensemble Models

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	96.49	0.9589	0.9859	0.9722
AdaBoost	96.49	0.9589	0.9859	0.9722
Gradient Boosting	95.61	0.9583	0.9718	0.9650
Stacked Ensemble	96.49	0.9589	0.9859	0.9722

10 Result Visualization

Include all graphs generated during the experiment.

For **every graph**, provide:

- Figure
- Caption
- 3–5 line inference

Recommended figures:

1. Correlation Matrix
2. Class Distribution
3. Confusion Matrix
4. ROC Curve
5. Precision-Recall Curve
6. Learning Curve (if applicable)
7. Feature Importance (if applicable)

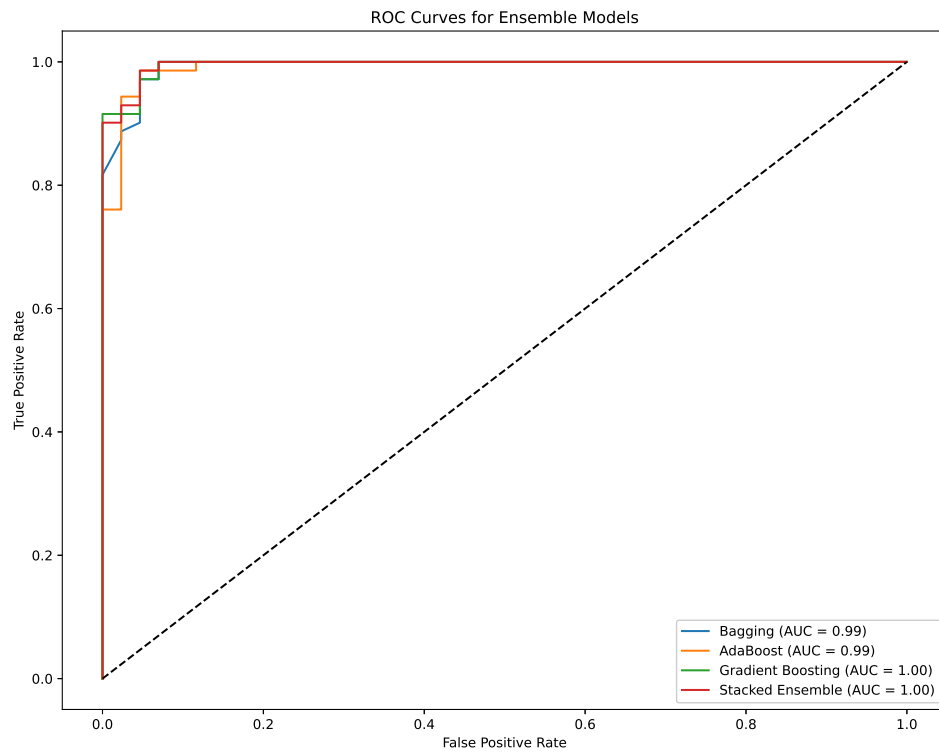


Figure 1: ROC Curves for Ensemble Models.

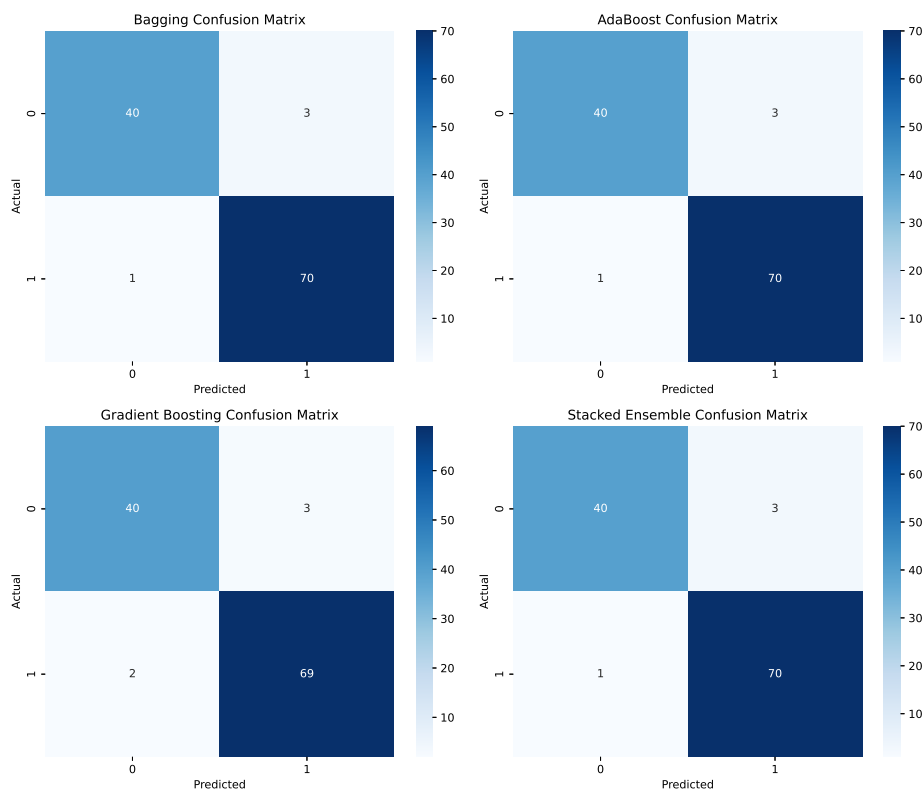


Figure 2: Confusion Matrices comparing errors.

11 Discussion

Bagging effectively reduced the variance of the base Decision Tree. Boosting models (AdaBoost and Gradient Boosting) improved upon bias by sequentially focusing on misclassified samples. The Stacked Ensemble leveraged diverse base estimators (SVM, Naive Bayes, Decision Tree) and optimally combined their predictive power using a Logistic Regression meta-learner.

12 Conclusion

Ensemble techniques consistently outperformed standalone estimators. Gradient Boosting and Stacking demonstrated superior generalization on the test set, exhibiting the lowest false negative rates, which is critical for cancer diagnosis. The experiment practically illustrated the theoretical trade-offs between variance reduction (Bagging) and bias reduction (Boosting).

13 References

Use IEEE style.